28	**D** The uncertainty principle states that the uncertainty in a particle's position, Δx and the uncertainty in its momentum, Δp are inversely related by the formula: $\Delta x \Delta p \gtrsim h$. This means that a particle with a highly certain (well-known) position will have a highly uncertain momentum, and vice versa. $\lvert\psi\rvert^2$ represents the probability density of finding the particle at a specific position x. The **narrower** the peak of the graph, the smaller the uncertainty in position, Δx because the particle is highly localized to a specific spot. **Graph A:** The probability is completely flat and spread out over all space. Position uncertainty, Δx is infinitely large, so momentum uncertainty Δp is at its minimum. **Graph B:** The probability oscillates across the entire range. The position is spread out over many peaks (Δx is large), indicating a well-defined momentum value (small Δp). **Graph C:** The particle is spread out over a wide wave packet spanning from x = 0.0 to 3.2. This represents a moderately large Δx. **Graph D:** The probability density shows a very sharp, narrow peak localized tightly around x = 1.5. This represents the smallest uncertainty in position Δx. Since Graph D has the smallest Δx, it must have the largest uncertainty in momentum Δp.
29	**A**